- Ass 3 due on Fri 9 Oct @ 4pm

Recap:

(1) A "hypothesis test" is simply a decision rule that specifies when to reject one claim H_0 in favour of another claim H_1

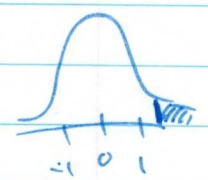
(2) The decision rule to reject H_0 in favour of H_1 if $\{T(\underline{X}) \in C\}$ has Type I error rate $\alpha = P_{H_0}(T(\underline{X}) \in C)$
↑ if H_0 true — incorrect rejection.

Example 7.1 (continued) In our medical study on lowering LDL cholesterol levels, we found that the typical patient-to-patient variability in cholesterol levels was $\sigma = 1.1$ mmol/L. Based on $n = 27$ samples in our study, the decision rule

> Reject H_0 in favour of H_1 if $\bar{X} \geq 0.3$ mmol/L

has Type I error rate

$$\alpha = P_{H_0}(\bar{X} \geq 0.3)$$
$$= P_{H_0}\left(\frac{\bar{X} - 0}{1.1/\sqrt{27}} \geq \frac{0.3 - 0}{1.1/\sqrt{27}}\right)$$
$$\simeq P(Z \geq 1.417)$$
$$= 7.8\%$$



Similarly, the decision rule

> Reject H_0 in favour of H_1 if $\bar{X} \geq 0.5$

has Type I error rate

$$\alpha = P_{H_0}(\bar{X} \geq 0.5)$$
$$\simeq 1\%$$

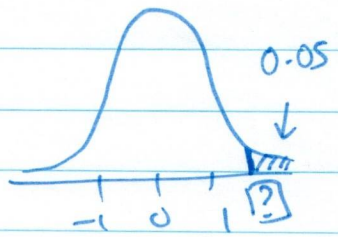
& the decision rule

> Reject H_0 in favour of H_1 if $\bar{X} \geq 0.2$

has Type I error rate

$$\alpha = P_{H_0}(\bar{X} \geq 0.2)$$
$$\simeq 17.2\%$$

Suppose we want to control the Type I error rate at 5%. Then what critical value should we use in our decision rule?



$$0.05 = P(Z \geq 1.645)$$

$$\overset{CLT}{\approx} P_{H_0}\left(\frac{\bar{X}-0}{1.1/\sqrt{27}} \geq 1.645\right)$$

$$= P_{H_0}(\bar{X} \geq 1.645 \times 1.1/\sqrt{27})$$

$$= P_{H_0}(\bar{X} \geq 0.348 \text{ mmol})$$

So here, a hypothesis test at level 5% would be

> Reject H_0 in favour of H_1 if $\bar{X} \geq 0.35$ mmol/L

It seems tempting to aim for a test with minimal (or zero) Type I error. But the only rule that achieves this is "always accept H_0". This rule is useless in practice because it lacks power — the ability to detect an effect if there is indeed an effect.

Type II error & power

Accidentally retain H_0 when it is in fact false, this is called a Type II error. The prob. of making a Type II error is

$$\beta = P(\underbrace{T(\underline{X}) \notin C}_{\text{accepting } H_0} \mid H_1)$$

$$= P_{H_1}(T(X) \notin C)$$

$$= 1 - P_{H_1}(T(\underline{X}) \in C)$$

($P_{H_1}(T(\underline{X}) \in C)$: prob of a correct rejection.)

By def, the power is $P_{H_1}(T(\underline{X}) \in C) = 1 - \beta$

Since H_1 is typically a composite hypothesis (e.g. $\mu > 0$) there are many values of parameter in the alternative space. For each value, we can compute the power of the test & then graph it as a fn of the parameter. This is called a power curve.

Example 7.1 (continued) In our medical study, deciding between $H_0: \mu = 0$ vs $H_1: \mu > 0$. We have seen that the test $\{\bar{X} \geq 0.35\}$ has Type I error rate of 5%. What is the power of this test?

power $= P_\mu(\bar{X} \geq 0.35)$ for each value of $\mu > 0$

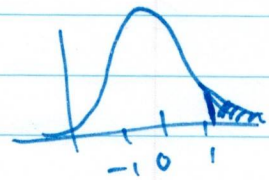
Let's suppose that the true is $\mu = 0.1$ mmol/L (& NOT $\mu = 0$)

Then power $= P_{\mu=0.1}(\bar{X} \geq 0.35)$

$$= P_{\mu=0.1}\left(\frac{\bar{X} - 0.1}{1.1/\sqrt{27}} \geq \frac{0.35 - 0.1}{1.1/\sqrt{27}}\right)$$

$$= P(Z \geq 1.181)$$

$$= 0.119 = 11.9\%$$



So the test $\{\bar{X} \geq 0.35\}$ will correctly detect an effective drug 11.9% of the time if the true effect of the drug is 0.1 mmol/L

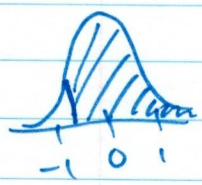
On the other hand, let's suppose the true effect is $\mu = 0.5$ mmol/L (& NOT $\mu = 0$), then

power $= P_{\mu=0.5}(\bar{X} \geq 0.35)$

$$= P\left(\frac{\bar{X} - 0.5}{1.1/\sqrt{27}} \geq \frac{0.35 - 0.5}{1.1/\sqrt{27}}\right)$$

$$= P(Z \geq -0.709)$$

$$= 76.1\%$$



So if the true effect of the drug is 0.5mmol/L, then our decision rule $\{\bar{X} \geq 0.35\}$ will correctly detect this 76.1% of the time

In general, if the true mean is μ, then the power is

$$\begin{aligned}
\text{power} &= P_\mu(\bar{X} \geq 0.35) \\
&= P\left(\frac{\bar{X}-\mu}{1.1/\sqrt{27}} \geq \frac{0.35-\mu}{1.1/\sqrt{27}}\right) \\
&= P\left(Z \geq \frac{0.35-\mu}{1.1/\sqrt{27}}\right) \\
&= 1 - \Phi\left(\frac{0.35-\mu}{1.1/\sqrt{27}}\right)
\end{aligned}$$

where Φ is the cdf of a standard normal.

We can use software to compute this for for a grid of μ values & plot it. R DEMO

We see that there is always a trade-off between Type I error & power. Neyman-Pearson proposed a compromise between Type I error & power. It is the most widely used paradigm today.

Neyman-Pearson paradigm: Aim for a test that:

1. restricts/controls the type I error to a "suitably low" level α (e.g. medicine $\alpha = 0.01$, general sciences $\alpha = 0.05$, social science $\alpha = 0.10$)
2. maximises the power under the alternative(s)

Any test that satisfies these two criteria is called a "most powerful test at level α". Neyman-Pearson (1933) proved that such tests can exist & provided a general approach to construct such tests via the likelihood ratio principle.

Likelihood ratio test

Let $L(\theta | \underline{X}) = f_\theta(\underline{X})$ be the likelihood function evaluated at θ for given dataset $\underline{X}$

Suppose (for now) that we are simply deciding between a single null $H_0: \theta = \theta_0$ versus another single alternative $H_1: \theta = \theta_1$, where θ_0 & θ_1 are two specified competing values.

Consider the likelihood ratio:

$$\frac{L(\theta_0 | \underline{X})}{L(\theta_1 | \underline{X})}$$

$L(\theta_0 | \underline{X})$ ← the likelihood of observing data $\underline{X}$ under H_0

$L(\theta_1 | \underline{X})$ ← the likelihood of observing data $\underline{X}$ under H_1

If this ratio is "small" $\Rightarrow$ data would have been more likely observed under H_1 (θ_1) than under H_0 (θ_0), so this would constitute evidence against H_0 in favor of H_1

This leads us to the Likelihood ratio test (LRT):

> Reject $H_0: \theta = \theta_0$ in favor of $H_1: \theta = \theta_1$ if $\frac{L(\theta_0 | \underline{X})}{L(\theta_1 | \underline{X})} \leq c$, for some critical value c

Interestingly, the LRT often simplifies to familiar tests that we have seen before! Moreover, the LRT is the LRT is the most powerful test possible. This result is due to Neyman-Pearson (1933):

Neyman-Pearson Lemma

Suppose the LRT that rejects $H_0: \theta = \theta_0$ in favour of $H_1: \theta = \theta_1$ if

$$\frac{L(\theta_0 | \underline{x})}{L(\theta_1 | \underline{x})} \leq c$$ such that it has level α

Then any other test with same level α must have power no larger than the LRT

Proof: see STAT2904 next week